Fading Voices

How Bird Song Travels Across the Amazonian Rainforest

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Abstract

Acoustic communication in dense environments like the Amazon rainforest is limited by signal degradation. In the lekking bird screaming piha, individual identity is encoded in song, yet the range of effective recognition remains unknown. Here we show that vocal signatures remain detectable for at least 100 meters and recognizable up to 120 meters. Using synthesized and real propagation data, classifiers achieved 85% accuracy in identifying individuals. Recognition was highest when training at 25 meters, matching typical lek spacing. These findings define the vocal active space of this species and reveal how acoustic constraints shape social communication.

1. Introduction

The screaming piha (Lipaugus vociferans), a bird inhabiting the dense Amazon rainforest, exhibits a lek-based social structure, forming small groups of 7 to 10 individuals that collectively occupy an area spanning 200 to 300 meters. Within these leks, each bird defends an individual territory, typically separated by 10 to 50 meters from its neighbors (Ulloa et al. (2016), Grison (2015)). Previous studies have established that these birds can recognize and distinguish their lek-mates based on vocalizations, with each individual carrying its identity in its song (Depriester (2019), Depriester (2021)). However, the dense rainforest environment poses significant challenges to signal transmission, as ambient noise and vegetation attenuate and degrade their vocal signals over distance. While the ability to identify its peers has been documented, the distance over which this recognition remains effective has never been assessed.

Here, we investigate how the screaming piha song degrades during propagation in order to assess the ”active space” of vocal individual signatures. First, we confirm the presence of distinct individual identity within original audio recordings of their songs. Next, we evaluate the performance of a classifier in distinguishing screaming piha vocalizations from background noise across increasing distances. Finally, we quantify the distance over which the identity encoded in these songs remains discernible, providing insight into the effective range of vocal recognition in this species within their acoustically complex habitat.

Figure 1: A Screaming Piha vocalizing in its natural habitat. Several dozen metres apart, the males recognize each other by their voices.

2. Experimental set-up

NE PAS RENDRE LE CODE et les data $->$ juste lien vers le github avec lien vers data

The dataset used in this study comes from a propagation experiment conducted within the natural habitat of the screaming piha, designed to closely reflect real-world conditions. The original propagation audio has been created from the vocalizations of 10 different individuals, with 10 different calls for each of them. The order of these 10 individuals within this audio band is predetermined and known, ensuring that the data are fully labeled. (Fig. 2)

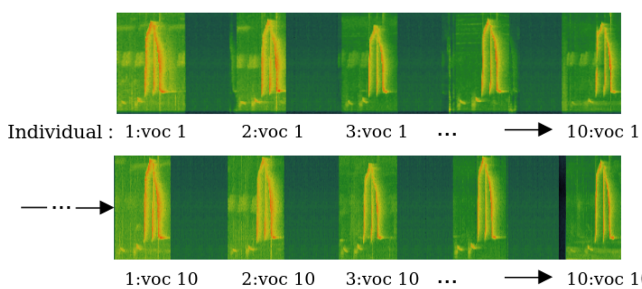


Figure 2: Original propagation band

The experiment consists in transmitting the original propagation band from a loudspeaker and recording it using a microphone placed at a certain distance. Four distances were tested: 12.5, 25, 50, 100 meters.

The dataset consists of 400 vocalization samples, comprising 10 calls from each of 10 individuals, recorded at four

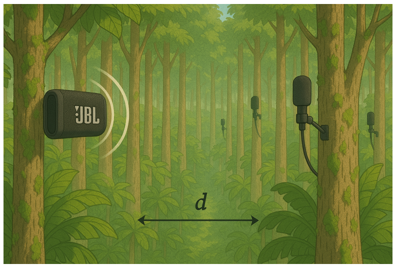


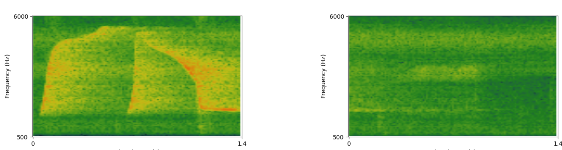
Figure 3: Experimental setup. $d \in [12.5, 25, 50, 100]$

different distances. To simulate realistic field conditions and ensure the classifier's ability to discriminate between signal and background, an additional 400 noise samples, each matched in duration and extracted immediately after each call, were included. This balanced design allows for the assessment of both vocalization detection and individual identification in the presence of natural ambient noise.

R 4.4.3 was used and the random seed is fixed at the start of each computation to get reproducible results.

3. Data preparation

Audio data come with an issue that we have to face before exploiting it. In fact, the WAV files are impossible to use as such. The data preparation/processing part is thus quite heavy.
To begin, we automatically extracted individual samples from each of the five propagation recordings (see songs_recup.ipynb). Representative spectrograms of an extracted vocalization and a background noise sample from the 12.5-meter recording are shown in Figures 4a and 4b, respectively.



(a) (b)

Figure 4: Spectrograms extracted from the 12.5 m. recording. (a) Vocalization. (b) Noise.

Following sample extraction, we selected a feature representation method suitable for our classification task. Given their ability to retain rich temporal and spectral information, we chose to use spectrograms. However, spectrograms are high-dimensional matrices that can be computationally expensive to process. To address this, we applied Principal Component Analysis (PCA) to reduce dimensionality while preserving the most information present the data. PCA transforms the original spectrogram features into a set of orthogonal components ranked by the variance they explain.
Because PCA results are sensitive to the input dataset, we used two distinct configurations: one combining vocalizations and background noise (DataVocNoise) to be used in signal-versus-noise discrimination, and another including only vocalizations (DataVoc) to emphasize inter-individual variation. The entire preprocessing workflow is illustrated in Figure 5.

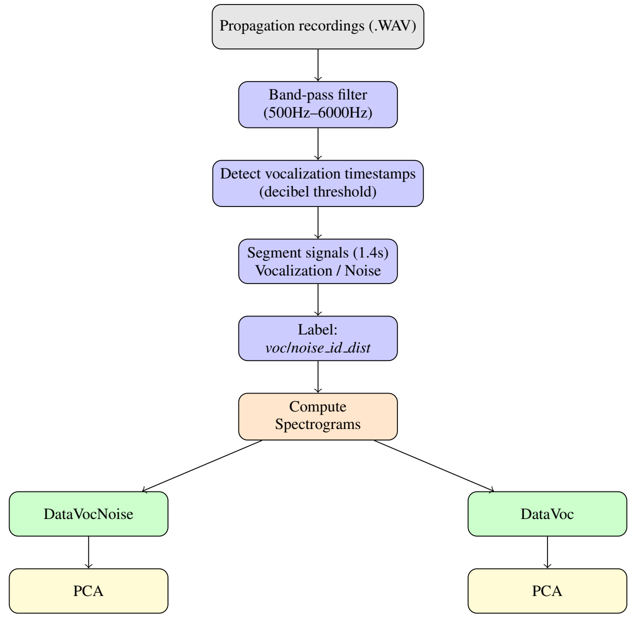


Figure 5: Compact view of the audio preprocessing and feature extraction pipeline.

4. Assessing Individual Acoustic Identity

Before investigating how vocal identity propagates across distance, it is necessary to verify that individual-specific acoustic information is indeed present in the vocalizations. To do so, we used a dataset of synthesized vocalizations, in which each signal was pre-processed to retain only its pitch contour.[1] These signals effectively isolate the core structure of each vocalization, excluding environmental background noise. A dataset is created by performing a PCA on the spectrograms of these synthesis signals (referred as DataSynth) . We will thus be able to assess, if the individual recognition works on DataSynth, that our classifier is using the vocalization and not the background noise to correctly classify individuals later on in DataVoc.

4.1. Choosing the right number of Principal Components

Although PCA reduces dimensionality, many components may still capture irrelevant variation. To assess information retention, we reconstructed spectrograms using 1, 20, and

[1]The pitch contour of a bird's vocalization refers to the temporal trajectory of its fundamental frequency, which often encodes species-specific and individual-level identity information relevant for communication and recognition.

50 components (Figure 6). The pitch contour is visible with just one component, but finer details required for individual recognition may be lost. Further analysis is therefore needed to select an appropriate number of components.

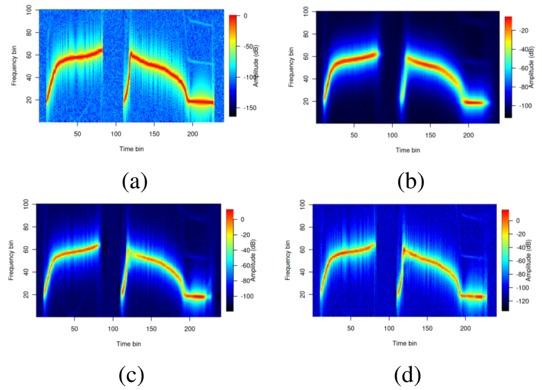


Figure 6: Spectrograms of a synthesized vocalization. (a) Original. (b) After reconstruction with 1 PC. (c) After reconstruction with 20 PC. (d) After reconstruction with 50 PC.

Figure 7 presents the mean quartic reconstruction error of the synthesized spectrograms as a function of the number of principal components. Finding the elbow point on this line helps us identify a potentially good number of PC, here around 19, that could capture most of the details of the spectrograms without capturing too much noise. We will therefore use 19 principal components in our futures analyses.

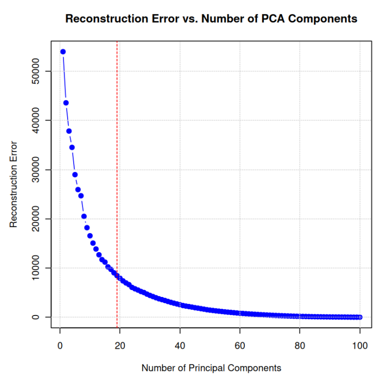


Figure 7: Reconstruction error w.r.t. the number of principal components. Red dashed line = 19 PC.

4.2. Choosing the right classifier

Using the DataSynth dataset, we performed supervised classification to assess individual acoustic identity. Selecting an appropriate classifier is critical: some models may be too simple to capture the complexity of the vocal signals, while others may be overly complex and require more data for effective training. Given that our dataset is relatively small (80 samples across 10 classes) and characterized by high dimensionality (19 features), we evaluated four candidate algorithms to identify the most suitable approach for our data.

- K-Nearest Neighbors: classifies a data point by finding the k closest points in the training set and assigning the most common label.
- SVM with Gaussian kernel: maps data into a higher-dimensional space using a radial basis function kernel, allowing it to find nonlinear decision boundaries.
- Random Forest: building multiple decision trees on random subsets of data and features and then combining their predictions.
- Multinomial log-linear model: extends logistic regression to predict probabilities for multiple categories using a softmax function.

To evaluate the performance of each classifier in identifying individuals, we applied a bootstrap resampling procedure (Algorithm 1).

Algorithm 1 Bootstrapping

Input: *df* dataframe of the vocalizations
Input: *B* number of bootstrap iterations
Input: *PC* number of principal components
Input: *model_algo* classification model
Input: *shuffleIndividuals* boolean
Output: metrics for each iteration
for iteration in range 1:B **do**
  $df\_bootstrap \leftarrow$ shuffle(df)
  if $shuffleIndividuals$ **then**
    $df\_bootstrap \leftarrow$ shuffle_labels($df\_bootstrap$)
  end if
  $X\_train, X\_test, y\_train, y\_test \leftarrow$ TrainTest($df\_bootstrap$)
  $model \leftarrow$ GridSearch_CrossVal(X_train, y_train)
  $trained\_model \leftarrow$ train(X_train, y_train)
  $metric\_iter \leftarrow$ test(X_test, y_test)
  $metrics[iter] \leftarrow metric\_iter$
end for
Return : metrics

This method provides a robust estimate of the model's generalization performance by repeatedly training and testing on random subsets of the data. By averaging results over many iterations, the bootstrap approach yields a statistically reliable approximation of the classifier's expected accuracy on unseen data.
The hyperparameter space of each model searched in function GridSearch_CrossVal is presented in table 1.

We applied the bootstrap procedure (Algorithm 1) to all four classifiers using $B = 100$ iterations and 19 principal components. The resulting mean classification accuracies are reported in Table 2. All models performed well above

Model	Hyperparameters
SVM (Radial)	$C \in \{1, 10, 50\}$ $\sigma \in \{0.01, 0.05, 0.1\}$
Random Forest	$mtry \in \{1, 2, 3, 5, 6, 7, 8, 9, 10\}$ $ntree = 1000$
KNN	$k \in \{1, 3, 5, 7, 9\}$
Multinomial Logistic Regression	$\emptyset$

Table 1: Hyperparameter grid for model selection

chance level, providing preliminary evidence that individual vocal signatures are reliably encoded in the synthesized vocalizations.

Table 2: Bootstrap Accuracy Results

Model	1st Qu.	Median	Mean	3rd Qu.
Random Forest	**0.800**	**0.833**	**0.834**	**0.867**
SVM (Radial)	0.750	0.800	0.805	0.854
Multinomial	0.746	0.800	0.783	0.833
KNN	0.700	0.750	0.755	0.808

Based on the results, the random forest classifier exhibited the highest performance and was thus selected for future analyses. We employed this algorithm in Sections 4.3 and 6, where the classification task remains focused on individual identification.

4.3. Assessing the individual recognition

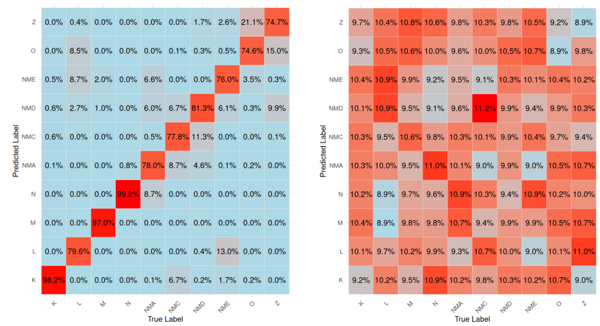
With the classifier selected, we next evaluated the strength of individual vocal signatures by testing whether classification accuracy remained consistently above chance. To do so, we compared two conditions: one in which the classifier was trained and tested on DataSynth with correct individual labels, and another in which the labels were randomly shuffled (DataSynthShuffled). This comparison allows us to assess whether performance is driven by meaningful individual differences in the vocalizations, which would indicate the presence of strong individual acoustic signatures. Results of the bootstrap evaluation (Algorithm 1) using $B = 1000$ iterations, 19 principal components, and the random forest algorithm are presented in Table 3.

Table 3: Bootstrap Accuracy Results using Random Forest

Dataset	1st Qu.	Median	Mean	3rd Qu.
DataSynth	0.800	0.833	0.833	0.867
DatSynthShuffled	0.050	0.100	0.096	0.133

The classifier trained and tested on DataSynth achieved an average accuracy of ~ 0.83, while the same classifier trained on DataSynthShuffled reached only ~ 0.10, equivalent to random guessing. This huge difference confirms that the classifier was able to reliably distinguish between individuals based on their vocalizations, providing evidence for the presence of individual acoustic signatures.

Figure 8 presents the confusion matrices for both datasets. Notably, certain individuals, such as individual M, were classified with near-perfect accuracy, suggesting particularly distinctive vocal signatures. This observation is further supported by Figures 9 and 10, which illustrate the clear separation of this individual in the acoustic feature space. Conversely, higher misclassification rates among some individuals may be explained by vocal similarity due to close genetic relatedness. As the screaming piha is a suboscine species[2], genetically similar individuals may produce more acoustically similar calls, contributing to reduced classification performance in those cases.



(a) (b)

Figure 8: Confusion Matrix of Classification (in % of correct classification). (a) DataSynth. (b) DataSynthShuffled.

Figure 9 illustrates the projection of DataSynth onto the first three principal components, while Figure 10 displays the decision boundaries of a random forest classifier trained on the first two components. Together, these visualizations provide insight into how a Bayes classifier[3] might perform on this dataset. Regions of class overlap—referred to as Bayes error regions[4]—are visible, confirming that some misclassifications are unavoidable. However, our full model was trained on 19 principal components, placing the data in a higher-dimensional space where more complex decision boundaries can be learned. This likely contributes to the classifier's strong performance, reaching an average accuracy of approximately 83%.

These results confirm the presence of individual acoustic signatures in the vocalizations of the screaming piha. Figure 11 illustrates an example of spectral differences observed between two individuals, highlighting the acoustic features that likely contribute to individual discrimination.

5. Distinguishing Vocalizations in the Rainforest's Acoustic Maze

The Amazonian rainforest, a dense and acoustically complex ecosystem with an extraordinary diversity of

[2]characterized by innate rather than learned vocalizations

[3]The Bayes classifier is the optimal theoretical classifier that achieves the lowest possible classification error given the true underlying data distribution.

[4]The Bayes error region is the subset of the input space where the class-conditional distributions overlap, leading to irreducible classification error.

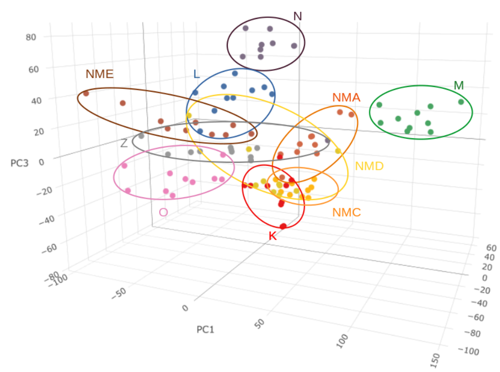


Figure 9: 3D Plot of DataSynth (accuracy of a random forest classifier trained and tested on 3PC : ~ 0.75

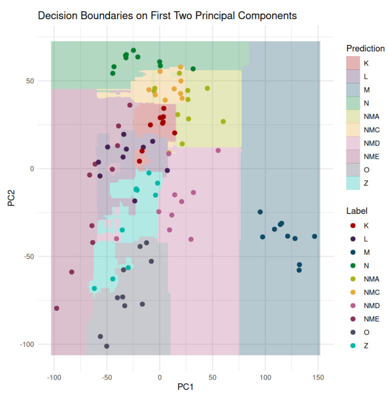


Figure 10: 2D Plot of DataSynth with decision boundaries of a random forest classifier trained on this dataset (accuracy of a random forest classifier trained and tested on 2PC : ~ 0.6)

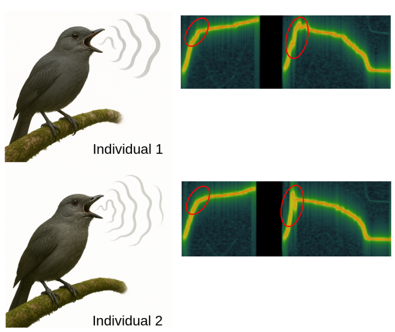
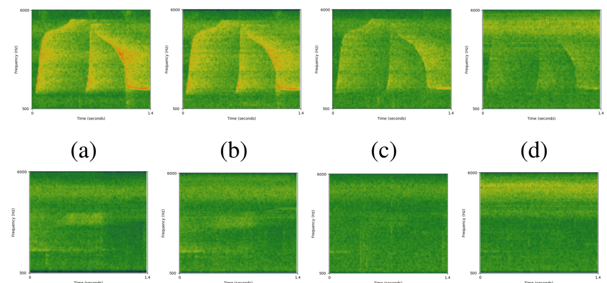


Figure 11: Highlight of the differences in the vocalizations of two different individuals.

vocalizing species, presents a challenging soundscape where the propagation of acoustic signals is hindered by dense vegetation, overlapping biological noises, and the constant interplay of competing calls. Before attempting to recognize individual identity, it is essential to determine the distance over which a screaming piha vocalization remains detectable. Building on the individual recognition demonstrated using synthesized data, we next evaluate whether screaming piha vocalizations can be reliably distinguished from background noise, and we estimate the maximum distance at which these vocalizations can still be detected within real propagation recordings.

To address this question, we used the DataVocNoise dataset, which includes both vocalizations and background noise, and framed the task as a binary classification problem: distinguishing between vocalizations and noise. Figure 12 illustrates representative spectrograms of vocalizations (top row) and corresponding background noise (bottom row) recorded at four distances—12.5, 25, 50, and 100 meters (left to right). As distance increases, the vocal signals progressively degrade and become increasingly difficult to distinguish from the ambient noise.



(a) (b) (c) (d)

(e) (f) (g) (h)

Figure 12: (a) Vocalization 12.5m. (b) Vocalization 25m. (c) Vocalization 50m. (d) Vocalization 100m. (e) Noise 12.5m. (f) Noise 25m. (g) Noise 50m. (h) Noise 100m.

Figure 13 shows the projection of the DataVocNoise dataset onto the first two principal components, with colors indicating vocalizations and noise. The separation between the two classes remains clear across all distances, suggesting that vocalizations retain distinct acoustic features that allow reliable discrimination from background noise, even as they propagate through the environment.

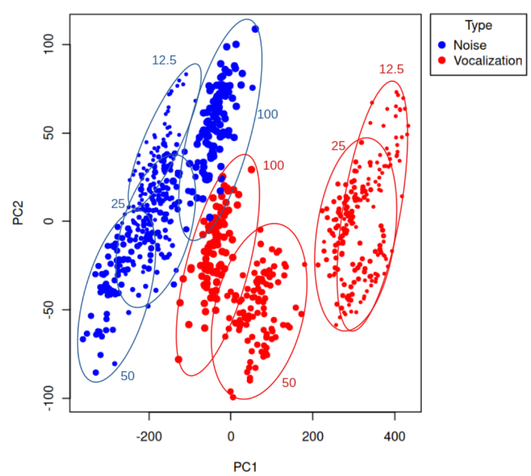


Figure 13: PCA of DataVocNoise

To evaluate the detectability of screaming piha vocalizations within propagation recordings, we trained a random forest classifier to distinguish between vocalizations and background noise using the DataVocNoise dataset. Figure 14 presents the classification accuracy over 100 training and testing iterations. The classifier achieved perfect accuracy at 12.5 and 25 meters, with very slight performance declines at 50 and 100 meters, yet still exceeding 98% accuracy in all cases. These results suggest that screaming piha vocalizations remain reliably detectable up to at least 100 meters. However, due to limitations in the available recordings, detection beyond this distance could not be assessed.

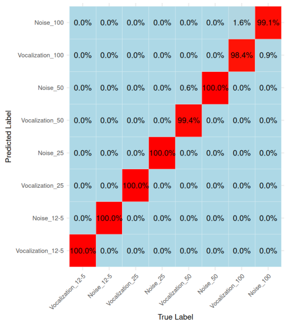


Figure 14: Confusion matrix of the random forest classifier trained and tested on DataVocNoise (in % of correct classification on the test set).

These results confirm that screaming piha vocalizations are detectable at distances of at least 100 meters. This means that, in the following section, any misclassification of individuals will not result from a failure to detect the vocalizations themselves, but rather from a loss or degradation of the individual vocal signature over distance.

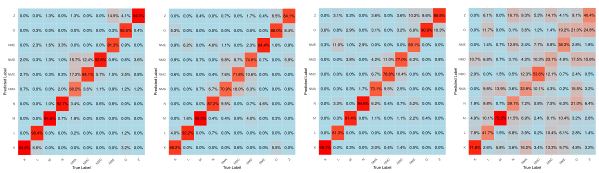
6. The Signature Within the Vocalization

In its natural habitat, the screaming piha is usually around 25 to 50 meters away from its neighbors. Consequently, individual recognition must often occur over these distances. Therefore, for assessing the recognition of other individuals we will use training sets composed of one or two distances at a time, and test on every distances. This design allows us to assess how the distance at which vocalizations are learned influences recognition performance at other distances. In doing so, we aim to identify the optimal learning distance for the screaming piha. That is, the distance at which individual identity in the vocalization is most robustly encoded and generalizable.

Classification accuracy was estimated using a bootstrap procedure (Algorithm 1) with $B = 100$ iterations and 19 principal components. Table 4 shows the results obtained after training on 25m recordings. We see that the mean accuracy on the three first distances is very close to the accuracy found in section 4.3, it is even a little bit higher which can be due to some loss of information in the synthesized signals. However, there is a sharp drop in accuracy at 100 meters, suggesting that individual vocal signatures degrade significantly beyond 50 meters. Figure 15 presents the corresponding confusion matrices. The same individuals that were difficult to classify in Section 4.3 remain challenging here, indicating that some vocal signatures are inherently harder to distinguish.

Table 4: Accuracy at Different Testing Distances after Training on 25 meters.

Testing Distance (m)	1st Qu.	Median	Mean	3rd Qu.
12.5	**0.817**	**0.866**	**0.852**	**0.900**
25	0.801	0.857	0.847	0.897
50	0.789	0.833	0.832	0.876
100	0.361	0.406	0.402	0.446



(a) (b) (c) (d)

Figure 15: Confusion Matrix of Classification after Training at 25m (in % of correct classification). (a) Testing on 12.5m. (b) Testing on 25m. (c) Testing on 50m. (d) Testing on 100m.

Figure 16 presents the results of the bootstrap analysis (with the same parameters as previously) across all training configurations. The results indicate that training the classifier on vocalizations recorded at 25 meters yields the highest overall recognition accuracy across all test distances. This suggests that 25 meters is the optimal distance for learning individual vocal signatures, as it enables reliable recognition not only at that distance but also at both shorter and greater ranges. Training on a combination of 25 and 50 meters also leads to high performance, although slightly lower than training on 25 meters alone. These findings are consistent with the typical spatial arrangement of screaming pihas within their leks, where individuals are generally spaced 20 to 40 meters apart. This may reflect a natural tuning of the vocal recognition system to the most ecologically relevant distances.

6.1. Vocal Active Space

Among all test distances, the lowest classification accuracy is observed at 100 meters. While performance at this distance is reduced, it still remains slightly above chance level (random guessing = 0.1, given ten individuals), suggesting that some residual identity information persists. To estimate the distance at which individual recognition becomes no better than random guessing, we fit a third-degree polynomial regression to the mean recognition accuracies obtained from the three best-performing training conditions: 25 meters, 50 meters, and the combination of 25 and 50 meters (see Figure 16).

The resulting regression curves are shown in Figure 17. All three models converge toward chance-level accuracy between approximately 115 and 125 meters, indicating a complete loss of individual-specific vocal information be-

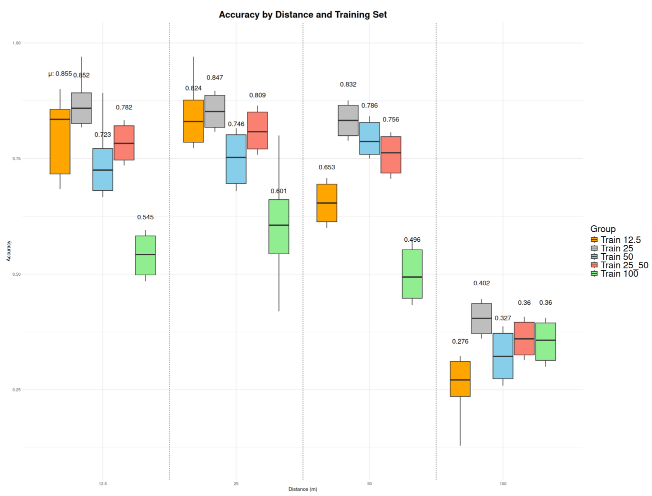


Figure 16: Accuracy Boxplots obtained from test set after training on different distances

yond this range. This suggests that, under natural conditions, screaming piha is likely unable to recognize known individuals at distances exceeding 120 meters, defining the upper boundary of the vocal active space of individual recognition for this species.

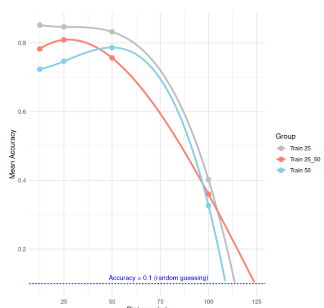


Figure 17: Regression curves showing the decline in recognition accuracy with distance for three training conditions. Chance-level accuracy is reached between 115 and 125 meters.

Figure 18 illustrates the estimated vocal active space for individual recognition in the screaming piha. This represents the spatial range, expressed in terms of recognition accuracy, within which a bird can identify the caller as a known neighbor, mate, or rival, based on individual-specific acoustic features embedded in its song. Beyond this range, environmental factors such as reverberation, attenuation, and background noise degrade the signal to the extent that recognition is no longer possible.

7. Conclusion

In this study, we investigated the ability of the screaming piha to convey and preserve individual identity through vocal signals in the acoustically complex environment of the Amazon rainforest. By combining signal processing, dimensionality reduction, and supervised classification, we con-

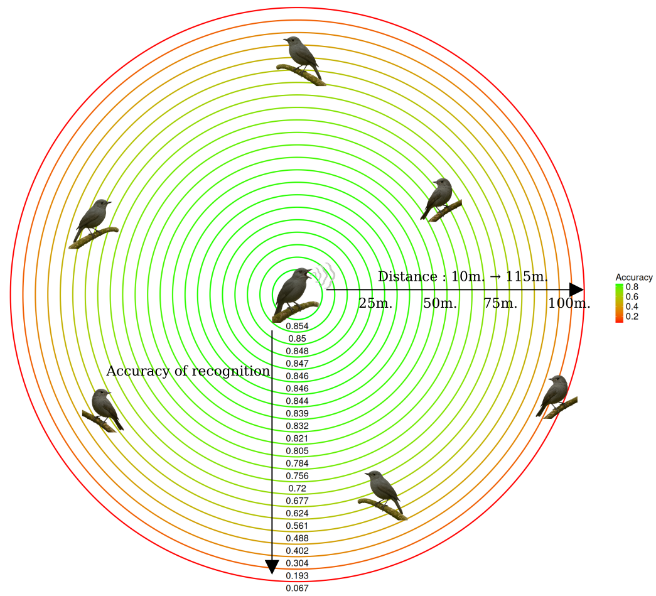


Figure 18: Vocal active space of the Screaming Piha. Vocal Accuracies where computed from the regression line obtained from the training at 25 meters.

firmed the presence of strong individual vocal signatures in this species. Classifiers trained on synthesized vocalizations reliably distinguished individuals, indicating that identity information is robustly encoded in the acoustic structure of their songs.

We then showed that L. vociferans vocalizations remain detectable up to at least 100 meters, with minimal loss in signal detectability. However, individual recognition begins to degrade with distance, particularly beyond 50 meters. Our analysis revealed that training on vocalizations recorded at 25 meters yields the best recognition performance across all test distances, suggesting that this range represents an optimal distance for identity learning, coinciding with the typical spacing of individuals within leks. Finally, by fitting polynomial models to recognition accuracy as a function of distance, we estimated that the upper limit of the vocal active space for individual recognition lies between 115 and 125 meters.

Together, these findings provide a detailed characterization of the spatial constraints on acoustic individual recognition in screaming piha, offering insight into how vocal communication systems operate in dense tropical environments. They also suggest that the spatial organization of leks in this species may reflect a compromise between signal detectability and identity recognition range, likely shaped by both ecological and evolutionary pressures.

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